Exsheet 3

1) $(\mathcal{L}u)(x) = -\sum_{i,j=1}^{d} a_{ij}(x)\, u_{x_i x_j}(x) + c(x)\, u(x) \leq 0 \;,\; x \in \Omega$

$c(x) \geq 0 \quad A(x) - \text{s.p.d} \quad \forall x \in \Omega$

$$\max_{x \in \bar{\Omega}} u(x) \leq \max\left\{0, \max_{x \in \Gamma} u(x)\right\}.$$

Proof: Assume

$$M := \max_{x \in \bar{\Omega}} u(x) > \max\{0, \max_{x \in \Gamma} u(x)\}$$

$M > 0$ and max is not on the boundary.

It means we take the maximum inside.

Let $y \in \Omega$. Then by hint, we have

$$\nabla u(y) = 0 \quad \text{and} \quad u_{\varepsilon_i \varepsilon_i}(y) \leq 0$$

Since A is symmetric positive definite, we can diagonalize $A(y)$ with orthogonal transformation.

$\Rightarrow A(y) = Q^T D Q$. Then

$$\underbrace{\sum a_{ij}(y)\, u_{x_i x_j}(y)}_{\leq 0} = \sum \underbrace{\lambda_i}_{>0} \underbrace{u_{\varepsilon_i \varepsilon_i}(y)}_{\leq 0}$$

$$(\mathcal{L}u)(y) = \underbrace{-\sum_{i,j=1}^{d} a_{ij}(y)\, u_{x_i x_j}(y)}_{\geq 0} + \underbrace{c(y)}_{\geq 0} \underbrace{u(y)}_{M > 0} \Rightarrow (\mathcal{L}u)(y) \geq 0.$$

But the given $(\mathcal{L}u)(x) \leq 0$. So, $(\mathcal{L}u)(y) = 0$

$$(\mathcal{L}u)(y) = 0 \Rightarrow \begin{cases} \sum_{i,j=1}^{d} a_{ij}\, u_{x_i x_j}(y) = 0 = \sum_{i=1}^{d} \underbrace{\lambda_i}_{>0} \underbrace{u_{\varepsilon_i \varepsilon_i}(y)}_{\leq 0} \\ \underbrace{c(y)}_{\geq 0} \underbrace{u(y)}_{>0} = 0 \Rightarrow c(y) \underset{\text{must}}{=} 0 \end{cases}$$

but needs to be 0.

$u_{\varepsilon_i \varepsilon_i}(y) = 0 \Rightarrow$ constant locally $\Rightarrow u$ cannot be isolated $\Rightarrow u$ - constant

Therefore, $\max_{x \in \bar{\Omega}} u(x) \leq \max\{0, \max_{x \in \Gamma} u(x)\}$.

2) $(\mathcal{L}u)(x) = u(x) - u''(x)$

a) $\Omega = (0,\pi)$, $u = -\sin x$

$u'' = \sin x$, $Lu = -\sin x - \sin x = -2\sin x \leq 0$ $(0,\pi)$

Boundary: $u(0) = 0$, $u(\pi) = 0 \Rightarrow \max_{\partial\Omega} u = 0$.

Interior maximum: $u(\pi/2) = -1 \Rightarrow \max_{\bar{\Omega}} u = 0$.

$\max_{\bar{\Omega}} u = 0 = \max\{0,0\}$.

b) $\Omega = (\pi/6, 5\pi/6)$, $u = 1 - \sin x$

$u'' = \sin x$, $Lu = (1-\sin x) - \sin x = 1 - 2\underbrace{\sin x}_{\geq \frac{1}{2}} \Rightarrow Lu \leq 0$

$$\left.\begin{aligned} u(\pi/6) &= 1 - 1/2 = 1/2 \\ u(5\pi/6) &= 1 - 1/2 = 1/2 \end{aligned}\right\} \Rightarrow \max_{\partial\Omega} u = 1/2$$

$u(\pi/2) = 0 \Rightarrow \max_{\bar{\Omega}} u = 1/2$

$\max_{\bar{\Omega}} u = 1/2 = \max\{0, \tfrac{1}{2}\}$.

c) If $c(x) < 0$, then

$$(\mathcal{L}u)(y) = -\sum_{i,j=1}^{d} a_{ij} u_{x_i x_j} + c(y)u(y)$$

Interior maximum: $-\sum_{i,j=1}^{d} a_{ij} u_{x_i x_j} \geq 0$ but $c(y)u(y) \overset{\text{could be}}{<} 0$

$\Rightarrow (\mathcal{L}u)(y) < 0$. It means it's allowed the maximum is interior.

The maximum principle fails.